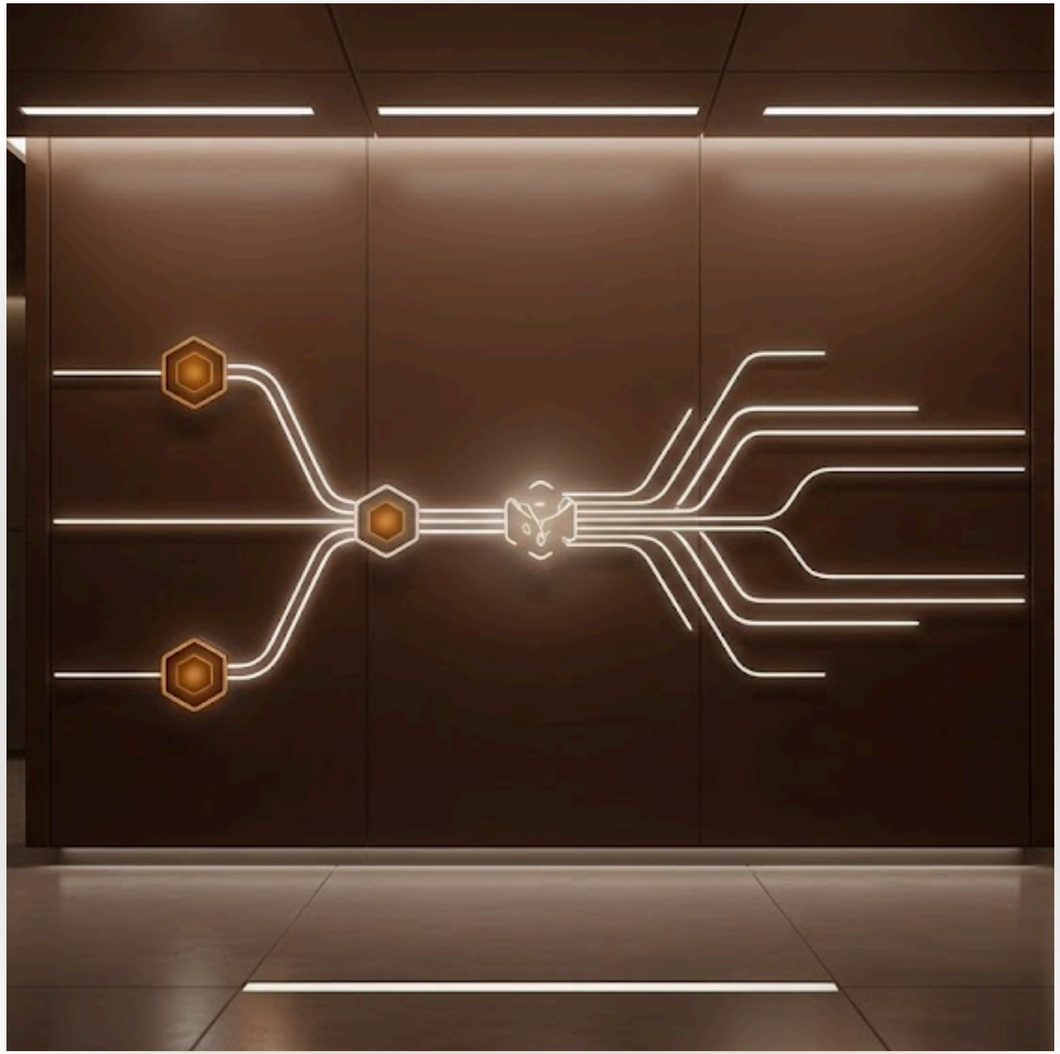


TRACK 04 · THE WORK BEHIND THE WORK

OpsPilot AI

Intelligent Operations Recovery & Business Impact Analysis Platform

Detect. Trace. Recover. Verify.

BEYOND BYTES

[Sania Mactelin A P · Sowndariya D]

Simulated pipeline data and controlled recovery.

PROBLEM STATEMENT

A pipeline failure is rarely just a pipeline failure.

Modern organizations rely on data pipelines, dashboards and reports for daily decisions. When a pipeline fails, runs late or delivers incomplete data, downstream assets and business teams can be affected. Monitoring often shows the technical error without making the business impact clear. Teams still have to trace dependencies, coordinate recovery and judge whether the data can be trusted.

The result: delayed decisions, and reliance on outdated or unreliable reports.



Pipeline failure → **Hidden downstream impact** → **Stale reports** → **Uncertain decisions**

EXISTING SOLUTIONS AND GAPS

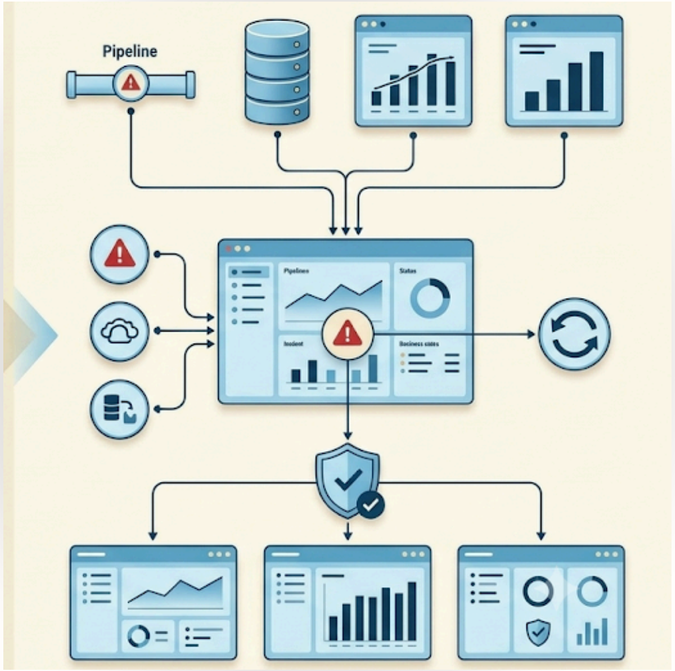
Monitoring tells us what broke. Teams still need to know what it affects.

Traditional monitoring

- Detects job failures and technical errors.
- Provides alerts and execution logs.
- May require manual investigation of downstream impact.

Manual recovery workflows

- Depend on operator investigation and coordination.
- Can involve repeated troubleshooting.
- May require separate checks to confirm data correctness.



The remaining gap

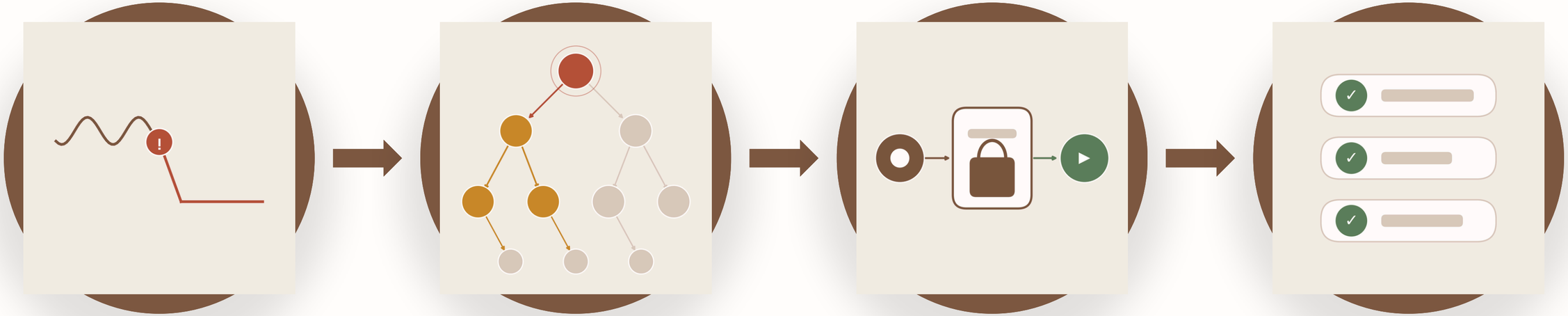
- Unified visibility into technical failure and business impact.
- Controlled recovery with explicit approval.
- Verification before incident resolution.
- Clear trust status for affected reports and dashboards.

OpsPilot AI is designed to address this gap.

Knowing a job failed is not the same as knowing which reports and teams are affected.

PROPOSED SOLUTION

One workflow from failure detection to verified recovery.



DETECT

Identify pipeline failures and data-quality issues.

TRACE

Identify downstream dashboards, reports, assets and business teams affected.

RECOVER

Recommend suitable actions and run predefined recovery procedures after required authorization.

VERIFY

Validate execution and data quality before resolving the incident.

A recovery attempt is not proof of recovery. Verification must pass before the incident can be resolved.

SYSTEM ARCHITECTURE

A connected architecture for safe, explainable operations recovery.

FRONTEND

React + Vite

- Operations dashboard
- Incident investigation
- Dependency graph
- Recovery approval interface
- Verification results

API

BACKEND

Python + FastAPI

- Pipeline monitoring and simulation
- Incident management
- Dependency traversal
- Business impact analysis
- Recovery orchestration
- Verification engine
- Audit logging

Read / write

DATABASE

SQLite

- Pipeline records and history
- Assets and dependencies
- Incidents and status history
- Recovery actions and approvals
- Verification results
- Audit events

CORE WORKFLOW



TECHNOLOGY STACK

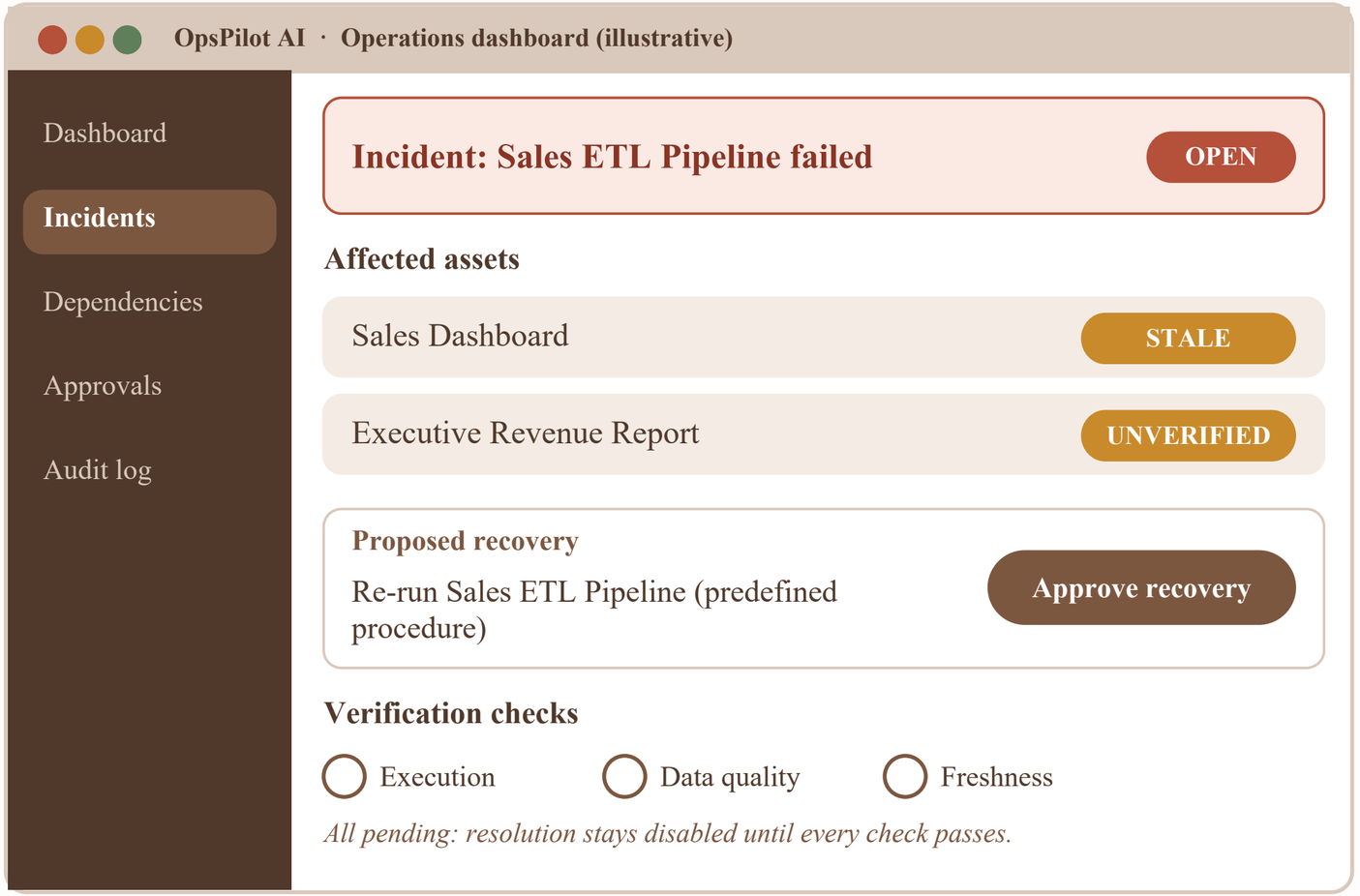
Practical technologies. A focused implementation.

1	**React + Vite**	Responsive, interactive frontend with rapid development.
2	**Python + FastAPI**	Backend APIs and modular operations logic.
3	**SQLite**	Lightweight relational persistence, suited to a hackathon prototype.
4	**Dependency graph**	Makes relationships and downstream impact easy to see.
5	**Rule-based logic**	Explainable decisions and testable recovery safeguards.



WORKING PROTOTYPE / DEMO

From a pipeline failure to a verified recovery.



Illustrative mockup, not a screenshot. Pipelines and recovery are simulated in the prototype.

1. Pipeline failure is triggered
2. System creates an incident
3. Impact analyzer finds downstream assets
4. Recovery action is proposed
5. Required approval is obtained
6. Recovery runs in the demo environment
7. Quality and freshness checks run
8. **Incident resolves only if verification passes**

Example: the Sales ETL Pipeline fails; the Sales Dashboard and Executive Revenue Report are marked stale or unverified until checks pass.

INNOVATION AND KEY DIFFERENTIATORS

From detecting failure to proving recovery.

1

Impact-aware operations

Connect technical pipeline failures to downstream business assets and affected teams.



2

Guarded recovery

Require authorization before executing protected recovery procedures.

3

Verification before resolution

Prevent incidents from being marked resolved merely because a retry or recovery action completed.

4

Trust-aware reporting

Keep affected dashboards and reports marked stale or unverified until required checks pass.

The innovation is the integration: one coherent workflow from detection to proven recovery.

FEASIBILITY AND SCALABILITY

Feasible to build now. Designed to evolve.

MVP · this prototype

- Lightweight, beginner-friendly stack.
- Simulated pipelines and controlled recovery actions.
- Relational persistence for incidents and audit history.
- Deterministic dependency traversal and verification rules.
- Runs locally, without expensive infrastructure.

Scalability path · future work, not built

1. Replace simulated execution with real orchestration integrations
2. Connect production data catalogs and lineage sources
3. Move to a more scalable database when required
4. Add role-based access control and stronger authentication
5. Add asynchronous job processing and observability
6. Extend verification policies and notification integrations

IMPACT AND BENEFITS

Better operational visibility. More trustworthy business decisions.

Operational

- Reduce manual investigation effort.
- Improve visibility into incident status.
- Standardize recovery procedures.

Technical

- Identify downstream dependencies.
- Preserve a traceable recovery history.
- Validate data quality after recovery.

Business

- Make affected reports visible.
- Reduce the risk of relying on stale data.
- Improve confidence in operational reporting.

Evaluation framework

- Pipeline failure detection rate
- Accuracy of downstream impact identification
- Recovery verification pass rate
- Incidents resolved without required verification
- Audit trail completeness

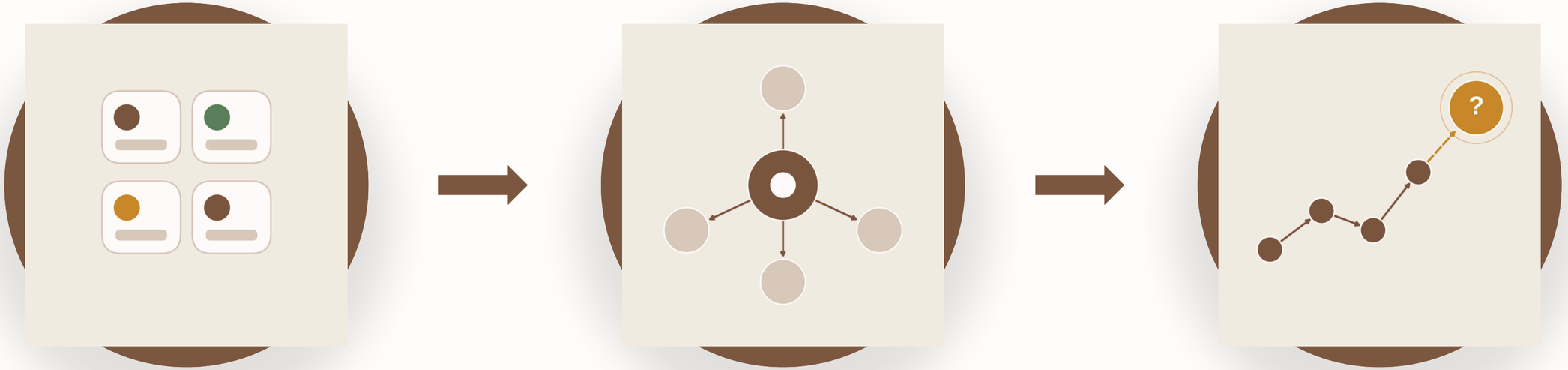
DESIGN TARGET, NOT A MEASURED RESULT

Zero incidents resolved without passing required verification checks.

FUTURE SCOPE

From a controlled prototype to connected operational intelligence.

Future possibilities, not current features



NEAR TERM

- Improve failure classification and recommendation rules.
- Expand data-quality checks.
- Enhance scenario simulation and reporting.
- Add configurable approval policies.

NEXT STAGE

- Integrate with real data orchestration platforms.
- Connect to production catalogs and lineage systems.
- Add notifications and team collaboration.
- Add authentication and role-based permissions.

LONG TERM

- Introduce predictive failure detection.
- Prioritize incidents using richer business context.
- Support multiple teams and workspaces.
- Explore explainable AI-assisted diagnosis.

Thank you

Don't just detect the failure.
Restore confidence.

OpsPilot AI

Detect. Trace. Recover. Verify.